

LESSON PLAN

Subject: Computer Science

Chapter: Programming Fundamentals

Topic: Algorithms and Flowcharts

Grade Level: 7

Duration: 45 minutes

Teacher: Abishek Khawas

Date: 2082/03/31

1. Learning Objectives

By the end of this lesson, students will be able to:

- Define what an **algorithm** and a **flowchart** are.
- Explain the importance of step-by-step instructions in programming.
- Draw a basic flowchart for a simple problem (e.g., adding two numbers).
- Use digital tools to create or visualise flowcharts.

2. Teaching Materials and ICT Tools Required

- Computer/Laptop with Internet
- Projector and whiteboard
- **Google Slides** for concept presentation
- **YouTube** (short video on flowcharts)
- **Lucid chart** or **Draw.io** (for creating flowcharts online)
- **Google Classroom** for assignment submission
- Printed handouts (for offline backup)

3. Lesson Plan

A. Introduction (5 minutes)

Objective: Activate prior knowledge and introduce algorithms.

Activity:

- Begin with a question: *“How do you make tea? What steps do you follow?”*
- Connect the idea to programming: *“Programming also follows steps, called algorithms.”*

B. Explanation & Demonstration (15 minutes)

Objective: Explain the concepts of algorithms and flowcharts.

ICT Tools: Google Slides, YouTube

Activities:

- Show a **Google Slides** presentation:
 - Define **Algorithm**: Step-by-step instructions to solve a problem.
 - Provide example:

1. Start
2. Input A and B
3. $\text{Sum} = A + B$
4. Display Sum
5. End

- Define **Flowchart**: A visual representation of an algorithm.
- Introduce key flowchart symbols:
 - Oval $\rightarrow$ Start/End
 - Parallelogram $\rightarrow$ Input/Output
 - Rectangle $\rightarrow$ Process
 - Diamond $\rightarrow$ Decision
- Play a **YouTube video** (2–3 minutes) to demonstrate flowcharts visually.

C. Main Activity (15 minutes)

Objective: Practice creating a basic flowchart.

Option A: Offline Activity

- Students draw a **flowchart on paper** for the task:
“Find the largest of two numbers.”

Option B: Online Activity

- Students use **Lucidchart** or **Draw.io** to create the same flowchart.
- Submit a screenshot or file through **Google Classroom**.

D. Assessment & Reflection (5 minutes)

Objective: Reflect on the learning and check understanding.

Activities:

- Ask:
 - “Why are algorithms important in programming?”
 - “How does a flowchart help you think clearly?”
- Provide oral or written feedback on student flowcharts.

E. Closure (5 minutes)

Objective: Reinforce learning and connect to real-life application.

ICT Tool: Whiteboard / Mind Map (offline)

Activity:

- Summarize with a **mind map** on the board:

Programming → Algorithm → Flowchart → Clear Instructions

- Encourage students to try making flowcharts for real-life tasks (e.g., brushing teeth, preparing lunch).

4. Homework/Extension Activities

- Draw a **flowchart** at home for one of the following:
 - Brushing your teeth
 - Preparing lunch
 - Going to school
- Write the **algorithm steps** for the same task.
- Optional: Use **Lucidchart** or **Draw.io** to create a digital version and submit via **Google Classroom**.